Md Zahidul Islam

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Professional Summary

Ph.D. candidate in Industrial Engineering (Operations Research) with experience in mathematical optimization, simulation, and data science. Applied these methods in research on workforce scheduling, supply chains, routing, and healthcare operations, and in graduate projects on forecasting and predictive modeling. Works primarily in Python and translates operational requirements and data into models that support decisions. Author of eight peer-reviewed publications.

Education

Clemson University — Clemson, South Carolina, USA
Ph.D. Candidate, Industrial Engineering (Operations Research); GPA: 3.90/4.00 — *Aug 2023 – Expected Dec 2027*
Advisor: Dr. Kevin M. Taaffe, Harriet and Jerry Dempsey Professor and Chair

University of Tennessee – Knoxville — Knoxville, Tennessee, USA
M.S. in Industrial Engineering; GPA: 4.00/4.00 — *Aug 2021 – Aug 2023*

Rajshahi University of Engineering & Technology (RUET) — Rajshahi, Bangladesh
B.Sc. in Industrial & Production Engineering; CGPA: 3.90/4.00, 1st in class — *Jan 2012 – Nov 2016*

Technical Skills

Programming: Python, SQL, R, MATLAB, C

Optimization: Mixed-Integer Linear Programming, Constraint Programming/CP-SAT, Multi-Objective Optimization, Lagrangian Relaxation, Branch and Bound, Dynamic Programming, Markov Decision Processes, Queueing Theory

Simulation & Solvers: Discrete-Event and Agent-Based Simulation; Gurobi, Google OR-Tools, CPLEX, AnyLogic, Arena

Machine Learning & Forecasting: Supervised Learning, Demand Forecasting, Neural Networks and LSTM, Gradient Boosting, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Deep Reinforcement Learning

Statistics: Regression Analysis, Hypothesis Testing, Design and Analysis of Experiments, Mixed-Effects Models

Python Libraries & Tools: NumPy, SciPy, pandas, scikit-learn, TensorFlow, PyTorch, XGBoost; Git, SLURM/HPC

Relevant Coursework

Integer Programming; Nonlinear Programming; Markov Decision Processes; Stochastic Processes; Probabilistic Methods for Industrial Engineering; Advanced Optimization via Simulation; Applied Data Science; Reinforcement Learning

Research Experience

Clemson University — Clemson, SC, USA
Graduate Research Assistant, Department of Industrial Engineering — *Jan 2025 – Present*

- Lead doctoral researcher on a federally funded, multi-institution scheduling project with Prisma Health, collaborating with faculty in industrial engineering, psychology, emergency medicine, and sleep science.
- Translated clinician and hospital requirements into multi-site workforce scheduling models (MILP and CP-SAT) covering staffing levels, eligibility, credentials, rest, fatigue, fairness, and individual preferences.
- Implemented the models in Python with Gurobi and Google OR-Tools and compared solution approaches for large instances on a high-performance computing cluster; schedules met an average of 84–87% of each physician's weighted preference categories across synthetic 25–75-physician benchmarks while satisfying all hard staffing rules.
- Formulated and solved a bi-objective model trading clinician fatigue against operational schedule quality and fairness for physicians rotating across multiple hospitals over a 90-day planning horizon.
- Integrated work schedules, surveys, interviews, wearable-device, and physiological data into analysis-ready datasets, and turned the findings into model inputs, constraints, and evaluation measures.
- Coordinate multi-site data collection and supervise undergraduate research assistants; present results to clinical collaborators to inform scheduling recommendations.

University of Tennessee Space Institute — Tullahoma, TN, USA
Graduate Research Assistant — *Aug 2021 – Jul 2022*

- Built an AnyLogic supply chain simulation (Winter Simulation Conference paper) in which combined information sharing and risk pooling reduced simulated order-to-demand variance by 46.3% at the factory and 49.3% at the wholesaler.
- Worked on a supervised-learning approach for the traveling salesman problem (IEEE T-ITS journal article) and a Gurobi production-scheduling MILP using plastics-manufacturer data (IISE Annual Conference presentation).

Selected Operations Research & Data Science Projects

Bike-Sharing Demand Forecasting *Graduate Class Project*

- Forecast hourly bike-rental demand in Seoul from weather, time-of-day, season, holiday, and operating-status features, comparing an artificial neural network, a PyTorch LSTM on sliding-window sequences, XGBoost, and random forest.
- Tuned the models with Ray Tune, staged grid search, and GridSearchCV, raising the neural network's test R^2 from 0.91 to approximately 0.96, and used correlation and feature-importance analysis to identify demand drivers.

Dynamic Physician Allocation using Markov Decision Processes *Graduate Class Project*

- Formulated sequential allocation of physicians between critical and mild-patient queues as a discounted MDP, uniformizing a continuous-time queueing model with class-specific arrivals, service rates, and holding costs.
- Solved it by dynamic programming with value iteration, policy iteration, and modified policy iteration, and by a Gurobi dual linear program, comparing runtime, convergence, value functions, and policy agreement.

Emergency Department Admission Prediction *Graduate Class Project*

- Built an end-to-end Python workflow on 347,360 MIMIC-IV-ED encounters, integrating relational tables and combining 50 structured predictors with 500 TF-IDF features from chief-complaint text.
- Compared logistic regression, random forest, XGBoost, CatBoost, and LightGBM with cross-validated grid search, evaluating ROC-AUC, precision, recall, and F1; CatBoost reached the best ROC-AUC (0.863).

Lagrangian Relaxation with Subgradient Multiplier Updates *Graduate Class Project*

- Relaxed time-indexed resource constraints into the objective with Lagrange multipliers updated by a subgradient method, tracking bounds and recovered feasible solutions.
- Analyzed step-size sensitivity and benchmarked the relaxation against the original formulation and an LP bound.

Uncapacitated Facility Location and Branch and Bound *Graduate Class Project*

- Implemented a custom branch-and-bound routine in MATLAB with Gurobi (fractional-variable branching, variable fixing, incumbent tracking) and benchmarked it against the integer solver and LP relaxation on 10–50-facility instances.

Learning-Based Optimization for the Traveling Salesman Problem *Peer-Reviewed Journal Publication*

- Co-authored research developing a supervised-learning approach that predicts structural components of high-quality TSP solutions and integrates them with exact and heuristic methods; published benchmarks showed a 28% smaller average optimality gap and 59% less computation time than a graph pointer network on 1,000-node instances.

Additional Professional Experience

Clemson University & University of Tennessee – Knoxville SC / TN, USA
Graduate Teaching Assistant *Aug 2022 – Dec 2024*
Supported courses in systems modeling, quality engineering, supply chain design, and engineering economics.

Rajshahi University of Engineering & Technology (RUET) Rajshahi, Bangladesh
Lecturer and Assistant Professor, Industrial & Production Engineering *Feb 2019 – Aug 2021*
Taught Probability and Statistics, Operations Management, and Industrial Simulation, and advised undergraduate research and capstone projects in optimization, simulation, scheduling, manufacturing, and supply chain systems.

Ahsanullah University of Science and Technology Dhaka, Bangladesh
Lecturer, Mechanical and Production Engineering *Apr 2017 – Dec 2018*
Taught Operations Management, Engineering Economy, and Manufacturing Processes.

Energypack Engineering Limited Dhaka, Bangladesh
Intern *Jun 2015 – Jul 2015*
Prepared a quality manual for transformer fabrication processes and conducted facility layout planning for production areas.

Leadership & Professional Service

President, Clemson University INFORMS Student Chapter *Aug 2025 – Sep 2026*
Organized invited seminars, professional-development, and outreach events; co-hosted the *Optimality Gap* podcast. The chapter received the INFORMS Magna Cum Laude award in 2025 (as Vice President) and 2026 (as President).

Undergraduate Research Mentor, Clemson Creative Inquiry *Fall 2025 – Present*
Mentor twelve undergraduates from industrial engineering, biological sciences, bioengineering, health science, and nursing in field data collection, data-quality checks, analysis, and research presentation.

Peer Reviewer: *Journal of Big Data*; *Scientific Reports*; *BMC Medical Imaging*

Selected Honors & Awards

Dr. Lloyd W. Crawford Fellowship, UT Space Institute (2021) | University Gold Medal, RUET (2019) | Prime Minister Gold Medal, Bangladesh (2018) | Clemson CECAS GSAB and GSG Travel Grants (2026)